Atrey Desai

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Education

2023–27 (exp. May 27) **University of Maryland, College Park**
B.S. in Computer Science (Honors), B.A. in Linguistics
Minor in Korean Studies
Advisors: Rachel Rudinger, Jordan Boyd-Graber, & Mohit Iyyer
Coursework: Natural Language Processing (G), Commonsense Reasoning (G), Mechanistic Interpretability (G)*, Machine Learning, Data Science, Parallel Computing, Computer Systems, Syntax, Semantics*, Pragmatics*, Phonetics, Psycholinguistics (* in progress)

Research Experience

2024– **Undergraduate Researcher**, CLIP Lab, University of Maryland, College Park
Advisors: Rachel Rudinger, Jordan Boyd-Graber, & Mohit Iyyer
a. AI agent feedback for long-form text generation
b. Calibrating human reliance and susceptibility to VQA model explanations
c. Artifacts and distractor quality in human-written vs. LLM-generated MCQs
d. Adversarial benchmarking of VLMs on out-of-context video misinformation
e. Causal interpretability of syntactic learning in data-constrained LMs

2024– **Visiting Researcher**, ACL2 Lab & NSF, The University of Texas at Arlington
Advisor: Kenny Zhu
Unsupervised techniques for feline language discovery and categorization
AniVoice-cat: 26,000+ annotated cat vocalizations and 57 discovered phones

2025–26 **Member of Technical Staff**, Learn Prompting (San Francisco, CA)
Ran HackAPrompt, the world's largest AI red-teaming hackathon
Robustness of AI safety judges to universal and multimodal CBRNE jailbreaks

2023–24 **Researcher**, FIRE Sustainability Analytics Lab, University of Maryland
Advisor: Thanicha Ruangmas
Climate policy frameworks based on automated OCR of 50,000+ 10-K filings

2020–23 **Researcher**, Reinforcement Learning at Brown Group, Brown University
Advisor: Michael Littman
Human-readable reward interfaces letting non-experts specify RL tasks

Publications

Google Scholar * – equal contribution

Under Submission

P1. **Atrey Desai**, Nishant Balepur & Rachel Rudinger. *Quick, Create a Distractor! Evaluating LLM Distractors for Multiple-Choice Benchmarks.* Under review at ACL Rolling Review (ARR).

Peer-reviewed Proceedings

C1. Tuan Dang*, **Atrey Desai***, Hridayesh Lekhak*, Tirza Panunto*, Lindsay Pike*, Theron Wang* & Kenny Zhu*. *A Preview of Computational Animal Linguistics*. ACM Computing Surveys.

C2. **Atrey Desai** & Sathvik Nair. *Filling in the Mechanisms: How do LMs Learn Filler-Gap Dependencies under Developmental Constraints?* ACL 2026 (Findings). ⋆ **Oral, TLS 2026**

C3. Nishant Balepur, Bhavya Rajasekaran, Jane Oh, Michael Xie, **Atrey Desai**, …, Jordan Boyd-Graber. *BenchMarker: An Education-Inspired Toolkit for Highlighting Flaws in Multiple-Choice Benchmarks*. ACL 2026.

C4. Nishant Balepur, **Atrey Desai** & Rachel Rudinger. *Test-Time Reasoners Are Strategic Multiple-Choice Test-Takers*. ACL 2026.

C5. **Atrey Desai**, Nishant Balepur & Rachel Rudinger. *Language Models Generate Multiple-Choice Questions with Artifacts*. MASC-SLL 2025.

C6. Chace Hayhurst, Hyojae Park, **Atrey Desai**, Suheidy De Los Santos & Michael Littman. *Reinforcement Learning As End-User Trigger-Action Programming*. IML Workshop @ AAAI 2022; RLDM 2022.

Selected Awards & Honors

2026 CRA NSF CISE Student Funding Program ($9,000)

2026 ACL D&I Award ($750)

2026 Department of Computer Science Professional Development Fund ($500)

2025 SPIRE Research Grant ($3,000)

2024 Omicron Delta Kappa Top 10 Freshman

2023–26 CMSC & ARHU Dean's List

2023 UMD President's Scholarship ($50,000)

2023 NMSC National Merit Scholarship ($4,000)

2023 Catherine Yang Scholarship ($1,000)

Talks

2026 **Filler-Gap Dependencies under Developmental Constraints in LMs**
Texas Linguistics Society (Feb 2026)

2026 **Filler-Gap Dependencies under Developmental Constraints in LMs**
UMD Computational Cognitive Science Reading Group (Feb 2026)

2024 **Adaptor Grammars and Neural Networks for Feline Lexical Discovery**
University of Maryland (Nov 2024)

2024 **Adaptor Grammars and Neural Networks for Feline Lexical Discovery**
University of Texas at Arlington (Jul 2024)

Service & Leadership

2026– **Organizer**, QANTA/EMM-QA (2026)

2025– **Reviewer**, *ACL/ACL Rolling Review (ARR; Jan 2026, May 2026), ICML EMM-QA (2026), NeurIPS IAEval (2026)

2025– **Boardmember**, University Research Advisory Board

2025–26 **Vice Chair**, Computer Science Department Council
Elected by 4,200+ CS undergraduates (nation's largest student body) and appointed by faculty.

2024–26 **Senior Member**, FIRE Student Leadership Council
Represented 1000+ peers, ran events & workshops, and reformed class curricula.

2024– **University Ambassador** (Department & College)

Mentoring

2026 **REU CAAR:** Nathan Nguyen, Nghia Truong, Yasmine Tajeddin

2025 **Office of Undergraduate Research:** Juan Cortés, Kemisola Benson, Vivian Akpala

2024 **Technica:** Savya Miriyala, Tanya Grover, Jessica Ononye, Nakshatra Hiray

2020–22 **Robotics Curriculum:** designed and led for MSET

Skills

Programming Python, Java, JavaScript/HTML/CSS, R, MATLAB

Frameworks Harbor, Inspect, HF Transformers, PyTorch, Selenium, BeautifulSoup

Tools Git, Docker, GCP, Google Vertex AI, VS Code

Languages English (Native), Gujarati (Native), Spanish (Intermediate), Korean (Beginner)

Forecasting **#1 global forecaster** on Manifold Markets (Aug 2026); Top-3 (Mar–July, Sept 2026); Top 8 all-time in AI
#2 automated bot forecaster on Manifold Markets (Sept-Oct 2026)